Properly colored paths in the union of two Hamiltonian paths: from a refuted half-bound to an $O(\sqrt{n}\,)$ collapse

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Abstract

For a permutation σ of $\{0, 1, \ldots, n-1\}$, let H_σ be the two-edge-colored multigraph on $\{0, \ldots, n-1\}$ whose blue edges form the natural-order Hamiltonian path and whose red edges form the Hamiltonian path induced by the value order of σ, and let $\rho(\sigma)$ denote the maximum number of vertices on a properly colored path of H_σ (a simple path whose consecutive edges alternate in color). Every two-edge-colored multigraph that is the union of two Hamiltonian paths on a common vertex set arises this way. We study $\rho_{\min}(n) := \min_\sigma \rho(\sigma)$, motivated by a relaxation, introduced here, of the matching-edge parameter that drives the Norin–Steiner–Thomassé–Wollan bound for Lovász's problem on Hamiltonian paths in vertex-transitive graphs: a linear lower bound on $\rho_{\min}(n)$ would transfer, through an exact conversion lemma, to a linear lower bound on that parameter.

We first show the natural guess is wrong. A block-reversal family supplies properly colored paths on $(n + 3)/2$ vertices and, together with exhaustive computation through $n = 14$, suggested the half-bound $\rho(\sigma) \geq \lceil (n + 3)/2 \rceil$; we exhibit an explicit permutation at $n = 18$ with $\rho = 10 < 11$ that refutes it. We then prove the much stronger statement that the half-bound fails by a polynomial factor:

$$\rho_{\min}(n) = O(\sqrt{n}\,), \qquad \text{explicitly} \qquad \rho_{\min}(n) \leq 8\sqrt{n} + 20 \ \text{ for all } n \geq 100,$$

so $\rho_{\min}(n)/n \to 0$ and there is no constant $c > 0$ forcing a properly colored path on cn vertices. This answers the main question of the relaxation in the negative; we emphasize that it closes only that route, not the underlying matching-edge question (Section 1.3).

The proof introduces an inflation (substitution) operation on permutations together with a per-permutation invariant, the *transit number* $M(\pi)$, which bounds how many vertices a properly colored path can collect while passing through an inflated copy of π. The main lemma is a complete classification of the constrained terminal-to-terminal paths (*admissible visits*) of an explicit gadget family: for every even $b \geq 6$, the permutation $T_b = (0,\ b{-}1,\ 2,\ 3,\ \ldots,\ b{-}2,\ 1)$ admits exactly five admissible visits, of sizes $1, 2, 2, 3, 3$; in particular $M(T_b) = 3$, independent of b. The classification turns on a parity obstruction in a long run of doubled edges. For the resulting permutations $\sigma_{a,b}$ on $n = ab$ vertices we prove

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the two-sided bounds $2a+2b-7 \le \rho(\sigma_{a,b}) \le 4a+2b$, so the family provably realizes the order $\sqrt{n}$ when $a \asymp b$. Whether $\rho_{\min}(n) = o(n)$ can be complemented by any superconstant lower bound remains open: even $\rho_{\min}(n) \to \infty$ is not known. A second, independent construction with the same conclusion is given in an appendix, and all computations are reproducible from the ancillary code and logs. The proofs are elementary and self-contained; the computations are independent verifications, not ingredients of the proofs.

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1 Introduction

Let S_n denote the set of bijections $\sigma\colon \{0,1,\dots,n-1\} \to \{0,1,\dots,n-1\}$. Associate with $\sigma \in S_n$ the two-edge-colored multigraph H_σ on vertex set $\{0,1,\dots,n-1\}$ with

- **P-edges** ("position", blue): $\{i, i+1\}$ for $0 \le i \le n-2$;
- **V-edges** ("value", red): $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ for $0 \le v \le n-2$.

Each color class is a Hamiltonian path. A vertex pair may carry both a P-edge and a V-edge; these are retained as *parallel edges of different colors* ("doubled pairs"). Conversely, the union of any two Hamiltonian paths on a common vertex set, with one path blue and the other red, is color-isomorphic to H_σ for some σ (identify the vertex set with $\{0,\dots,n-1\}$ along the blue path, and let σ rank the vertices along the red path).

A *properly colored path* (PC path) is a simple path—all vertices distinct—whose consecutive edges have different colors. A single vertex is a PC path. Let $\rho(\sigma)$ be the maximum number of vertices on a PC path in H_σ, and

$$\rho_{\min}(n) \;=\; \min_{\sigma \in S_n} \rho(\sigma).$$

A useful picture. Plot σ as the point set $\{(i,\sigma(i)) : 0 \le i \le n-1\}$ in the plane. A P-edge joins horizontally adjacent columns; a V-edge joins vertically adjacent rows. A PC path traces an alternating horizontal/vertical staircase through this grid. Both color classes have maximum degree 2, so H_σ has maximum degree 4; this places the problem outside the regimes of standard properly-colored-path results, which typically require minimum degree linear in n [6].

1.1 Motivation: a relaxation of a matching-edge parameter

Lovász [1] asked in 1969 whether every connected vertex-transitive graph on n vertices contains a Hamiltonian path. The best general bound is $\Omega(n^{9/14})$, due to Norin, Steiner, Thomassé and Wollan [2], and rests on a reduction that depends on the matching-edge content of long simple paths in a family of graphs $G(n,\sigma)$ on $2n$ vertices, indexed by $\sigma \in S_n$. Writing

$$f(n) \;=\; \min_{\sigma \in S_n} \; \max_{\pi} \; \mu(\pi),$$

where π ranges over simple paths in $G(n,\sigma)$ and $\mu(\pi)$ counts the matching edges of π, the current bound $f(n) = \Omega(n^{4/5})$ comes from the circumference theorem of Liu, Yu and Zhang [3] on 3-connected cubic graphs, as used in [2]; replacing it with $f(n) = \Omega(n)$ would yield $\Omega(n^{2/3})$ on the Lovász problem. We define $G(n, \sigma)$ precisely in Section 3 and prove there a one-directional *conversion lemma*: a PC path on k vertices in H_σ yields a simple path in $G(n, \sigma)$ using k matching edges, so $f(n) \geq \rho_{\min}(n)$. A linear lower bound on $\rho_{\min}$ would thus give a linear lower bound on f. The relaxation in this direction is the route we introduce, and the main theorem closes it. The scope of that statement is delimited in Section 1.3.

Beyond this specific motivation, the question is a natural extremal problem about alternating paths. Properly colored ("alternating") paths and cycles in edge-colored graphs have been studied extensively since the survey of Bang-Jensen and Gutin [4]; see also [5, 6]. The positive results in that line are sufficient conditions—typically large color degree or dense hosts—for long PC paths or cycles to exist. The graphs H_σ sit at the opposite end of the spectrum: every vertex has at most two edges of each color, and the question is extremal rather than Dirac-type. We are aware of no prior work on the union of two Hamiltonian paths in this guise.

1.2 Results

The clean self-contained question, for a reader who sets the Lovász application aside, is:

> *Does there exist $c > 0$ such that every two-edge-colored multigraph that is the union of two Hamiltonian paths on the same vertex set contains a properly colored simple path on at least cn vertices?*

We answer it in the negative, and quantitatively.

Theorem 1.1 (Main Theorem)**.** *$\rho_{\min}(n) = O(\sqrt{n})$. Explicitly, $\rho_{\min}(n) \leq 8\sqrt{n} + 20$ for all $n \geq 100$. Consequently $\rho_{\min}(n)/n \to 0$.*

The route to this theorem passes through a refutation. Small cases and a locally extremal family pointed at a linear lower bound with constant 1/2:

> *Half-bound (refuted, Section 4):* for every $\sigma \in S_n$, $\rho(\sigma) \geq \lceil (n + 3)/2 \rceil$.

This holds on a block-reversal family σ_m (which attains it exactly) and through exhaustive enumeration of S_n for $n \leq 14$, but it is false: at $n = 18$ an explicit permutation σ^* has $\rho(\sigma^*) = 10 < 11$ (Theorem 4.6). The construction of σ^* – deleting one singleton from σ_4 to fuse two 3-blocks – is the seed of the gadget idea, and the main theorem shows that the half-bound fails not by an additive constant but by a polynomial factor.

The construction behind the main theorem is explicit. For even $b \geq 6$ let

$$T_b \;=\; (0,\ b{-}1,\ 2,\ 3,\ \ldots,\ b{-}2,\ 1) \in S_b$$

be the identity permutation with the values 1 and $b{-}1$ exchanged, and for $a \geq 2$ define $\sigma_{a,b} \in S_{ab}$ by

$$\sigma_{a,b}(jb + r) \;=\; jb + T_b(r) \qquad (0 \leq j < a,\ 0 \leq r < b),$$

i.e. a consecutive copies of T_b on consecutive value ranges. We prove the two-sided estimate

$$2a + 2b - 7 \;\leq\; \rho(\sigma_{a,b}) \;\leq\; 4a + 2b \qquad (a \geq 2,\ b \geq 6 \text{ even}), \tag{1}$$

the upper bound by the inflation calculus below (Theorem 8.1, Proposition 8.2), the lower bound by an explicit path (Proposition 8.3). Thus $\rho(\sigma_{a,b}) = \Theta(a + b)$: with $a \asymp b \asymp \sqrt{n}$ the family realizes the order $\sqrt{n}$ exactly, so the upper bound is tight for this family and not an artifact of a construction whose true value is smaller. (For $a = 2$ both bounds in (1) are weak; the construction is informative for $a \geq 3$, and asymptotically for $a \to \infty$.) Exact machine computation gives $\rho(\sigma_{a,b}) = 2a + 2b - 4$ at every tested point up to $n = 720$ (Section 9); we use this exact formula nowhere in the proofs and state it only as data.

The proof of the upper bounds has two independent components.

1. An inflation calculus (Sections 5–6). Call a set B of consecutive positions whose σ-image is a set of consecutive values an *interval block*. A short argument (Lemma 5.2) shows that at most *four* edges of H_σ join B to its complement, and that they attach at four canonical *terminal* vertices of B, with prescribed colors. If $\sigma = \tau[\pi_1, \ldots, \pi_a]$ is built by inflation (each position block an interval block with pattern π_i, arranged by an outer permutation τ), then any PC path decomposes into at most $2a - 1$ *visits* to blocks, and each visit other than the first and last must enter and leave through terminals subject to color constraints. Defining the *transit number* $M(\pi)$ as the largest number of vertices such a constrained passage (*admissible visit*) through H_π can collect, we obtain (Theorem 6.4)

$$\rho\big(\tau[\pi_1, \ldots, \pi_a]\big) \;\leq\; (2a-3)\,\max_i M(\pi_i) \;+\; 2\,\max_i |\pi_i|.$$

2. A classification of the gadget's admissible visits (Section 7). For every even $b \geq 6$ we determine the admissible visits of H_{T_b} *completely* (Theorem 7.4): up to reversal there are exactly five, of sizes $1, 2, 2, 3, 3$, so $M(T_b) = 3$, independent of b. The mechanism is a parity obstruction: H_{T_b} contains a long run of doubled edges which any terminal-to-terminal passage must either avoid or cross completely, and a full crossing forces the color of the departing edge precisely so that, for even b, every continuation is blocked. For odd b the parity reverses and a transit can collect all b vertices; exact computation confirms $M(T_b) = b$ for odd b. The even/odd contrast shows that the parity, not some feature of the argument, is what bounds the transit number.

Combining the two components and balancing a against b yields Theorem 1.1; the crude bound $\rho(\sigma_{a,b}) \leq 6a + 2b$, which needs only the value $M(T_b) = 3$ and not the full classification, already gives $\rho_{\min}(n) \leq 10\sqrt{n} + 10$, and the refined constant $8\sqrt{n} + 20$ uses the classification once more (Proposition 8.2).

Appendix A contains a *second, independent* construction proving $\rho_{\min}(n) = O(\sqrt{n})$: chains of reversed odd blocks separated by singletons, where a different parity trap blocks every crossing, with its own complete proof. The two constructions share the parity-trap idea but are different families with different edge inventories, and they were found and verified independently, by different methods and different solvers. We give both because a redundant proof of the main claim is worth the space.

Section 10 isolates what is known about lower bounds, including the machine-verified refutation of a natural conjectured inequality ($\rho \geq$ number of ± 1-blocks) that holds exhaustively through $n = 12$ but fails at $n = 240$.

1.3 Scope: what this does and does not say about Lovász's problem

The conversion lemma of Section 3 maps PC paths in H_σ to matching-edge-rich paths in $G(n, \sigma)$ *one-directionally*. Our construction makes properly colored paths short; it says nothing about paths in $G(n, \sigma)$ that are allowed to use runs of same-colored edges. The known bound $f(n) =$

$\Omega(n^{4/5})$, which rests on the circumference theorem of [3] as used in [2], is untouched, and $f(n) = \Theta(n)$ remains entirely possible. What dies is only the hope of proving $f(n) = \Omega(n)$ via the properly-colored relaxation introduced here.

On the lower-bound side our knowledge is thin: we prove no superconstant lower bound on $\rho_{\min}(n)$, and to our knowledge even $\rho_{\min}(n) \to \infty$ is open. The quantities controlling the natural approaches are discussed in Section 10, where we also state the conjecture $\rho_{\min}(n) = \Theta(\sqrt{n})$ — at present an upper bound plus structural circumstantial evidence, no more.

2 Preliminaries

Throughout, $\sigma \in S_n$ and H_σ are as in Section 1; colors are P and V, and for a color c we write $\bar{c}$ for the other color. Paths are sequences of distinct vertices $u_1, \ldots, u_k$ ($k \geq 1$) together with, for each consecutive pair, a chosen edge (with its color) joining them; in a doubled pair the path commits to one of the two parallel edges. A path is *properly colored* (PC) if consecutive edges have different colors. The *size* of a path is its number of vertices k; a PC path on k vertices has $k-1$ edges. Since each vertex pair carries at most one edge of each color and paths have distinct vertices, a path uses pairwise distinct edges, even in the multigraph sense.

Observation 2.1. A two-edge-colored multigraph is the union of a blue and a red Hamiltonian path on a common vertex set of size n if and only if it is color-isomorphic to H_σ for some $\sigma \in S_n$. Hence Theorem 1.1 is equivalent to its abstract formulation.

Remark 2.2 (Two notions of block)**.** Two block notions recur, and we keep them distinct. An *interval block* (Definition 5.1) is any set of consecutive positions whose values are consecutive; it carries an arbitrary *pattern*, and is the unit of the inflation calculus. A ± 1-*block* (Definition 4.1) is the special case in which consecutive values differ by a fixed ± 1; these are intrinsic to σ, partition the positions, and are the unit of the lower-bound discussion (Section 10). Every ± 1-block is an interval block; the converse fails.

3 The Conversion Lemma

This section makes precise the link to the matching-edge parameter of Section 1.1. It is logically independent of the rest of the paper and may be skipped by a reader interested only in $\rho_{\min}(n)$.

Definition 3.1. For $\sigma \in S_n$, let $G(n, \sigma)$ be the graph on $2n$ vertices $\{v_0, \ldots, v_{n-1}, u_0, \ldots, u_{n-1}\}$ with edges: a *top path* $v_0 v_1 \cdots v_{n-1}$; a *bottom path* $u_0 u_1 \cdots u_{n-1}$; and a *matching* $M = \{v_i u_{\sigma(i)} : 0 \leq i \leq n-1\}$. Every v_i and every u_j is incident to exactly one matching edge; the four path endpoints have degree 2 and every other vertex degree 3. For a simple path π in $G(n, \sigma)$, $\mu(\pi)$ is the number of edges of π in M, and $f(n) = \min_\sigma \max_\pi \mu(\pi)$.

Theorem 3.2 (Conversion Lemma)**.** *If H_σ has a PC path on k vertices, then $G(n, \sigma)$ has a simple path on $2k$ vertices using exactly k matching edges. Hence* $\max_\pi \mu(\pi) \geq \rho(\sigma)$ *for every σ, and $f(n) \geq \rho_{\min}(n)$.*

Proof. Let $w_0 w_1 \cdots w_{k-1}$ be a PC path in H_σ with edge colors $c_0, \ldots, c_{k-2}$ alternating. Build a path Π in $G(n, \sigma)$ on the $2k$ vertices $\{v_{w_i}\} \cup \{u_{\sigma(w_i)}\}$ as follows. If $c_0 = \mathrm{P}$, run

$$u_{\sigma(w_0)} \xrightarrow{M} v_{w_0} \xrightarrow{P} v_{w_1} \xrightarrow{M} u_{\sigma(w_1)} \xrightarrow{Q} u_{\sigma(w_2)} \xrightarrow{M} v_{w_2} \xrightarrow{P} v_{w_3} \xrightarrow{M} \cdots$$

alternating matching edges with top/bottom path edges; if $c_0 = \mathrm{V}$, start instead at v_{w_0}, symmetrically.

Edge validity. A P-edge $\{w_i, w_{i+1}\}$ means $|w_i - w_{i+1}| = 1$, so $v_{w_i}v_{w_{i+1}}$ is a top-path edge; a V-edge $\{w_i, w_{i+1}\}$ means $|\sigma(w_i) - \sigma(w_{i+1})| = 1$, so $u_{\sigma(w_i)}u_{\sigma(w_{i+1})}$ is a bottom-path edge. The color alternation of the PC path matches the alternation of top versus bottom traversal in Π, so every step of Π is a genuine edge.

Simplicity. The top vertices $\{v_{w_i}\}$ are distinct because the w_i are; the bottom vertices $\{u_{\sigma(w_i)}\}$ are distinct because σ is a bijection; the two families are disjoint.

Matching count. Π uses one matching edge per vertex of the PC path, namely $v_{w_i}u_{\sigma(w_i)}$ for $0 \le i < k$: a total of k.

Taking $\pi = \Pi$ gives $\max_\pi \mu(\pi) \ge \rho(\sigma)$ for each σ; the minimum over σ gives $f(n) \ge \rho_{\min}(n)$. $\square$

The conversion is one-directional: a path in $G(n, \sigma)$ that uses a run of two same-colored path edges (top or bottom) has no PC-path preimage, so the inequality $f(n) \ge \rho_{\min}(n)$ cannot be reversed. This is the precise sense in which our negative result for $\rho_{\min}$ does not transfer to f (Section 1.3).

4 The half-bound and its refutation

This section records the false start that motivates the rest of the paper. We exhibit a family on which ρ equals $\lceil (n+3)/2 \rceil$, state the half-bound it suggested, and then refute that half-bound by an explicit permutation at $n = 18$. The lower-bound construction and the refutation are proved in full; the exact value of the family is stated as computed data and is not used anywhere downstream.

Definition 4.1 (± 1-block)**.** A ± 1-*block* of σ is a maximal interval of positions $[p_\ell, p_r]$ such that either $p_\ell = p_r$ (a *singleton*) or there is a fixed $\varepsilon \in \{+1, -1\}$ with $\sigma(i+1) - \sigma(i) = \varepsilon$ for all $p_\ell \le i < p_r$. The ± 1-blocks partition $\{0, \ldots, n-1\}$; write $b(\sigma)$ for their number and $s_{\max}(\sigma)$ for the largest size.

Every internal edge of a ± 1-block is doubled (consecutive positions hold consecutive values), so a ± 1-block can be traversed in order with freely chosen alternating colors. This single fact gives two unconditional bounds.

Proposition 4.2 (Largest-block and one-vertex bounds)**.** *For every $\sigma \in S_n$, $\rho(\sigma) \ge s_{\max}(\sigma)$; and if $b(\sigma) \ge 2$, then $\rho(\sigma) \ge s_{\max}(\sigma) + 1$.*

Proof. Let B be a ± 1-block of size $s_{\max}$. Its internal edges are doubled, so traversing B in order with alternating $\mathrm{P}, \mathrm{V}, \ldots$ is a PC path on $s_{\max}$ vertices: this gives the first bound.

For the second, if $s_{\max} = 1$ then all blocks are singletons; with $b \ge 2$ the P-edge $\{0, 1\}$ exists and is a PC path on $2 = s_{\max} + 1$ vertices. Otherwise $s_{\max} \ge 2$. Since $b \ge 2$, a largest block B has a ± 1-block neighbor, hence an external edge e of some color c at a port $p \in \{p_\ell, p_r\}$ (Lemma 5.2 below gives the port structure for interval blocks, of which B is one). Traverse B from the port opposite p toward p, choosing the starting color so that the last internal edge has color $\ne \bar{c}$; then e, of color c, may be appended at p, yielding a PC path on $s_{\max} + 1$ vertices. $\square$

These are weak in absolute terms – the gadget family below has $s_{\max} = b$ fixed while $n \to \infty$ – but they are the floor any combined bound must clear, and they are the only unconditional lower bounds we have (Section 10).

4.1 The block-reversal family and the half-bound

Definition 4.3. For $m \geq 1$ and $n = 4m+3$, partition $\{0, \ldots, n-1\}$ into consecutive blocks of sizes

$$\big(2,\ \underbrace{3,1,3,1,\ldots,3,1,3}_{m \text{ threes, } m-1 \text{ singletons}}\ ,\ 2\big),$$

and let σ_m be the permutation that reverses each block (the values of each block run downward, with consecutive blocks carrying consecutive value ranges). For $m = 1$ this is the composition $(2,3,2)$ with no interior singleton. Explicitly,

$$\begin{aligned}
\sigma_1 &= (1,0,4,3,2,6,5),\\
\sigma_2 &= (1,0,4,3,2,5,8,7,6,10,9),\\
\sigma_3 &= (1,0,4,3,2,5,8,7,6,9,12,11,10,14,13),\\
\sigma_4 &= (1,0,4,3,2,5,8,7,6,9,12,11,10,13,16,15,14,18,17).
\end{aligned}$$

Each block of σ_m is an interval block (its values are consecutive), so σ_m is an inflation in the sense of Section 6; this is the viewpoint that the gadget construction later sharpens.

Proposition 4.4 (An explicit long path)**.** *For every $m \geq 1$, $\rho(\sigma_m) \geq 2m+3 = (n+3)/2$.*

Proof. Define a vertex sequence Γ_m by starting with $(4m-2,\ 4m-1,\ 4m)$; then for $j = m, m-1, \ldots, 2$ appending the pair $(4j-3,\ 4j-4)$; and finally appending $(0,1)$. For $m = 4$ this is $\Gamma_4 = (14,15,16,13,12,9,8,5,4,0,1)$, of length $3+2(m-1)+2 = 2m+3$. We check Γ_m is a PC path in H_{σ_m}.

The edges fall into four groups, whose colors are forced by Definition 4.3 (direct from the position/value adjacencies):

- the internal edges $(4m-2, 4m-1)$, $(4m-1, 4m)$ and $(0,1)$ are doubled (each lies inside a ± 1-block), hence freely colorable;
- each edge $(4j, 4j-3)$, $j = m, \ldots, 2$, joins the top of a 3-block to the adjacent singleton across a value-gap of 1 but a position-gap of 3: V-only;
- each edge $(4j-3, 4j-4)$, $j = m, \ldots, 2$, joins a singleton to the next 3-block across a position-gap of 1: P-only;
- the final edge $(4,0)$ joins the leftmost 3-block to the leftmost 2-block: V-only.

Reading Γ_m in order, the forced colors are $\mathrm{V}, \mathrm{P}, \mathrm{V}, \mathrm{P}, \ldots, \mathrm{V}, \mathrm{P}$ on the $2m+2$ edges, the doubled internal edges taking whichever color the alternation requires. All vertices are distinct, so Γ_m is a PC path on $2m+3$ vertices. □

Exhaustive computation through $n \leq 14$ found $\rho_{\min}(n) = \lceil (n+3)/2 \rceil$ tight at $n = 7, 8, 11, 12$ and slack by 1 elsewhere, with σ_m realizing the bound at $n = 4m+3$. Together with Proposition 4.4 this suggested:

Conjecture 4.5 (Half-bound; *false*)**.** *For every $n \geq 2$ and every $\sigma \in S_n$, $\rho(\sigma) \geq \lceil (n+3)/2 \rceil$.*

Exact computation gives $\rho(\sigma_m) = 2m + 3$ (so the lower bound of Proposition 4.4 is exact), verified for $m = 1, 2, 3, 4$ by independent solvers (Section 9); we record this as data, not as a theorem, and use it nowhere below. The reason we do not prove the matching upper bound $\rho(\sigma_m) \le 2m + 3$ here is deliberate: the inflation calculus of Section 6 gives only the weaker $\rho(\sigma_m) \le O(m)$ with a larger constant (each 3-block has transit number 3, so the generic bound does not see the singleton-flanking rigidity that pins the value to $2m + 3$), and the sharper argument would expand the paper without serving the main theorem, which makes the family entirely obsolete in Section 7. The role of σ_m here is only to motivate Conjecture 4.5 – and to be refuted.

4.2 An $n = 18$ counterexample

Theorem 4.6. *Let* $\sigma^* = (1, 0, 4, 3, 2, 5, 8, 7, 6, 11, 10, 9, 12, 15, 14, 13, 17, 16) \in S_{18}$*. Then* $\rho(\sigma^*) = 10 < 11 = \lceil (18 + 3)/2 \rceil$*, so Conjecture 4.5 is false at* $n = 18$*.*

The ± 1-block structure of σ^* has composition $(2, 3, 1, 3, 3, 1, 3, 2)$, i.e.

$$B_1 = \{0, 1\},\ B_2 = \{2, 3, 4\},\ B_3 = \{5\},\ B_4 = \{6, 7, 8\},\ B_5 = \{9, 10, 11\},\ B_6 = \{12\},\ B_7 = \{13, 14, 15\},\ B_8 = \{16,$$

Proof. Lower bound ($\rho \ge 10$), proved by exhibiting a path. The sequence

$$\Pi^* = (13, 14, 15, 12, 11, 6, 5, 2, 1, 0)$$

is a PC path on 10 vertices in H_{σ^*}. We verify each edge from the definition of σ^*: $\{13, 14\}, \{14, 15\}$ are doubled (internal to B_7); $\{12, 15\}$ is V-only (σ^{*-1} sends the consecutive values $\sigma^*(12), \sigma^*(15)$ to adjacent ranks, while the positions differ by 3); $\{11, 12\}$ is P-only; $\{6, 11\}$ is V-only (here $\sigma^{*-1}(8) = 6$ and $\sigma^{*-1}(9) = 11$); $\{5, 6\}$ is P-only; $\{2, 5\}$ is V-only; $\{1, 2\}$ is P-only; $\{0, 1\}$ is doubled. Assigning the colors V, P, V, P, V, P, V, P, V to the nine edges (the two doubled edges take the colors the alternation requires) gives a valid PC traversal, so $\rho(\sigma^*) \ge 10$.

Upper bound ($\rho \le 10$), by exhaustive search. An exact longest-PC-path search (state space: triples of visited set, current vertex, last color; every reachable state visited once, no pruning – Section 9) returns $\rho(\sigma^*) = 10$. Two independently written solvers (Python and C++) agree on the value, and the search at $n = 18$ for this single instance is small. Since $\lceil (18 + 3)/2 \rceil = 11$, the half-bound fails. □

Remark 4.7 (How σ^* was found, and why it matters)**.** σ^* is a single-position deletion from $\sigma_4 \in S_{19}$: delete the centre singleton (position 9, value 9) and relabel the values above it down by 1. Removing the singleton fuses the two flanking 3-blocks across a non-± 1 gap, changing the composition from $(2, 3, 1, 3, 1, 3, 1, 3, 2)$ to $(2, 3, 1, 3, 3, 1, 3, 2)$ and dropping $b(\sigma)$ from 9 to 8, while the deficit $d = n - \rho$ stays at 8. A *fused* pair of reversed 3-blocks across a non-± 1 gap is exactly a small instance of the parity trap that the gadget T_b exploits at scale: the deletion that breaks the half-bound is the seed of Theorem 1.1. (The values $n = 15, 16, 17$ have not been checked exhaustively, so we do not claim $n = 18$ is the smallest counterexample value.)

4.3 Structural results that survive

Besides the unconditional Proposition 4.2, one conditional comparison survives the refutation and is used nowhere below but is recorded for completeness. For σ with ± 1-blocks $B_1, \ldots, B_{b(\sigma)}$ in position order, the *block-quotient permutation* τ_σ ranks the blocks by their value range; H_{τ_σ} is the corresponding graph on $b(\sigma)$ vertices.

Proposition 4.8 (Odd-only quotient lift)**.** *If every* ± 1*-block of* σ *that is neither first nor last in position order has odd size, then* $\rho(\sigma) \geq \rho(\tau_\sigma)$*.*

Proof. Take a maximum PC path Q in H_{τ_σ} and lift it to H_σ by replacing each block-vertex visit by a traversal of the corresponding block. A block visited as a through-vertex of Q is entered and left by edges of different colors (the PC condition at the block-vertex); lifting it to a full traversal of an interval block of size s preserves a color difference between entry and exit if and only if s is odd (the $s-1$ internal edges alternate, so entry and exit colors differ iff $s-1$ is even). By hypothesis every internal block has odd size, so every such lift is consistent; the end blocks carry only one external edge and impose no parity constraint. The lifted path is a PC path of size $\geq \rho(\tau_\sigma)$. □

The hypothesis is met by σ_m (internal blocks of sizes 3 and 1). The naive lift fails for general σ: a quotient path demanding a full traversal of an *even* internal block forces equal entry/exit colors. The unconditional inequality $\rho(\sigma) \geq \rho(\tau_\sigma)$, verified empirically for $n \leq 11$, is open.

5 Interval blocks and ports

We now develop the machinery for the main theorem. The unit is the interval block of Remark 2.2.

Definition 5.1 (Interval block, pattern)**.** An *interval block* of $\sigma \in S_n$ is a set of positions $B = \{p_\ell, p_\ell + 1, \dots, p_r\}$ whose image is a set of consecutive values $\sigma(B) = \{w_\ell, \dots, w_r\}$. Its *pattern* is the permutation $\pi \in S_b$, $b = |B|$, defined by $\pi(t) = \sigma(p_\ell + t) - w_\ell$.

Lemma 5.2 (Port structure)**.** *Let* B *be an interval block of* σ *with positions* $[p_\ell, p_r]$*, values* $[w_\ell, w_r]$*, and pattern* π*. Then:*

(a) The sub-multigraph of H_σ *induced on* B *is color-isomorphic to* H_π *under the map* $p \mapsto p - p_\ell$*.*

(b) Every edge of H_σ *with exactly one endpoint in* B *is one of at most four: the P-edges* $\{p_\ell - 1, p_\ell\}$ *and* $\{p_r, p_r + 1\}$ *(when they exist), whose endpoints in* B *are* p_ℓ *and* p_r*; and the V-edges* $\{\sigma^{-1}(w_\ell - 1), \sigma^{-1}(w_\ell)\}$ *and* $\{\sigma^{-1}(w_r), \sigma^{-1}(w_r + 1)\}$ *(when they exist), whose endpoints in* B *are* $\sigma^{-1}(w_\ell)$ *and* $\sigma^{-1}(w_r)$*.*

Proof. (a) P-edges internal to B are the pairs $\{p, p+1\} \subseteq [p_\ell, p_r]$, which correspond exactly to the P-edges of H_π. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ has both endpoints in B iff both $v, v+1 \in [w_\ell, w_r]$ (as $\sigma(B) = [w_\ell, w_r]$ and σ is a bijection), and these correspond exactly to the V-edges of H_π under the stated map, since $\pi^{-1}(v - w_\ell) = \sigma^{-1}(v) - p_\ell$ for $v \in [w_\ell, w_r]$. Doubled pairs match doubled pairs, so the isomorphism preserves colors.

(b) A P-edge $\{p, p+1\}$ with exactly one endpoint in $[p_\ell, p_r]$ satisfies $p + 1 = p_\ell$ or $p = p_r$. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ with exactly one endpoint in B has exactly one of $v, v+1$ in $[w_\ell, w_r]$, forcing $v + 1 = w_\ell$ or $v = w_r$. □

Definition 5.3 (Terminals, roles, port colors)**.** Let $\pi \in S_b$, $b \geq 2$. The *terminals* of H_π are the four *roles*

$$p_0 = \text{vertex } 0, \quad p_1 = \text{vertex } b-1, \quad v_0 = \text{vertex } \pi^{-1}(0), \quad v_1 = \text{vertex } \pi^{-1}(b-1),$$

with *port colors* $\mathrm{pc}(p_0) = \mathrm{pc}(p_1) = \mathrm{P}$ and $\mathrm{pc}(v_0) = \mathrm{pc}(v_1) = \mathrm{V}$. Distinct roles may occupy the same vertex (a role is a label, not a vertex); since $b \geq 2$, two roles with the *same* port color always occupy distinct vertices.

By Lemma 5.2, in any interval block with pattern π the at most four outgoing edges attach (inside the block) precisely at the terminals, and the color of an outgoing edge equals the port color of the role at which it attaches.

Definition 5.4 (Admissible visit, transit number)**.** Let $\pi \in S_b$, $b \geq 2$. An *admissible visit* of H_π is either

(i) a PC path $u_1, \ldots, u_k$ in H_π with $k \geq 2$, together with a choice of roles r at u_1 and r' at u_k, such that the first edge has color $\neq \mathrm{pc}(r)$ and the last edge has color $\neq \mathrm{pc}(r')$; or

(ii) a single vertex carrying two distinct roles r, r' with $\mathrm{pc}(r) \neq \mathrm{pc}(r')$.

The *transit number* $M(\pi)$ is the maximum size of an admissible visit, and $M(\pi) = 0$ if none exists. For $b \leq 1$ set $M(\pi) = 1$.

The colour constraints encode that a visit is entered and left by external edges of the roles' port colours (which differ from the first and last internal edges by the PC condition). Two trivial but load-bearing facts:

Lemma 5.5 (Reversal; endpoint constraint)**.** (a) *If $(u_1, \ldots, u_k; r, r')$ is an admissible visit, so is the reversed $(u_k, \ldots, u_1; r', r)$; admissible visits may thus be classified as undirected vertex sequences with unordered role pairs.* (b) *In a type-(i) admissible visit, both u_1 and u_k are terminal vertices, the roles r, r' are distinct, and the conditions read: first-edge color $= \overline{\mathrm{pc}(r)}$, last-edge color $= \overline{\mathrm{pc}(r')}$.*

Proof. (a) Reversal swaps first and last edges and the two endpoint constraints. (b) The roles occupy $u_1 \neq u_k$, hence are distinct; the rest is the definition together with the fact that with two colors "$\neq \mathrm{pc}(r)$" means "$= \overline{\mathrm{pc}(r)}$". ☐

6 Inflation and the Inflation Lemma

Definition 6.1 (Inflation)**.** Let $a \geq 2$, let τ be a permutation of $\{1, \ldots, a\}$, and let $\pi_1, \ldots, \pi_a$ be permutations of sizes $b_1, \ldots, b_a \geq 2$, $N = \sum_i b_i$. Put $s_i = \sum_{j<i} b_j$ (position offsets) and $t_i = \sum_{j:\ \tau(j)<\tau(i)} b_j$ (value offsets). The *inflation* $\sigma = \tau[\pi_1, \ldots, \pi_a] \in S_N$ is

$$\sigma(s_i + r) = t_i + \pi_i(r), \qquad i = 1, \ldots, a, \quad 0 \leq r < b_i.$$

The value intervals $[t_i, t_i + b_i - 1]$ partition $\{0, \ldots, N-1\}$, so σ is a bijection. Each position block $B_i = [s_i, s_i + b_i - 1]$ is an interval block of σ with value interval $[t_i, t_i + b_i - 1]$ and pattern π_i, so Lemma 5.2 applies to all blocks simultaneously. Edges of H_σ with endpoints in different blocks are *cross edges*.

Lemma 6.2. *$H_{\tau[\pi_1,\ldots,\pi_a]}$ has exactly $2(a-1)$ cross edges, counted with multiplicity: $a-1$ P-colored (one per consecutive pair of position blocks) and $a-1$ V-colored (one per consecutive pair of value intervals). Each block is incident to at most 4 cross edges, attaching at its terminals as in Lemma 5.2(b), and each terminal role of each block carries at most one cross edge.*

Proof. A P-edge $\{p, p+1\}$ is a cross edge iff $p+1 \in \{s_2, \dots, s_a\}$: $a-1$ edges. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ is a cross edge iff $v, v+1$ lie in different value intervals, which happens for exactly the $a-1$ values v ending a value interval. The attachment statement is Lemma 5.2(b); per block, each of the four port positions is met by at most one cross edge. $\square$

Lemma 6.3 (Visit decomposition). *Let $\sigma = \tau[\pi_1, \dots, \pi_a]$, $a \geq 2$, and let Q be a PC path in H_σ. Partition the vertex sequence of Q into maximal runs of consecutive vertices lying in a common block; call these runs* visits. *Then:*

(a) Consecutive visits are joined by exactly one cross edge; distinct consecutive pairs use distinct cross edges. Hence if Q has q visits, then $q - 1 \leq 2(a-1)$, i.e. $q \leq 2a - 1$.

(b) Every visit that is neither the first nor the last is an admissible visit of the copy of H_{π_i} on its block; in particular it has at most $M(\pi_i)$ vertices.

(c) The first and the last visit have at most $\max_i b_i$ vertices each.

Proof. (a) The edge of Q joining the last vertex of one visit to the first vertex of the next has endpoints in different blocks (maximality of runs), so it is a cross edge. Q has distinct vertices, hence pairwise distinct edges, so the $q-1$ connecting edges are distinct cross edges; there are $2(a-1)$ of them in all (Lemma 6.2).

(b) Let $W = u_1, \dots, u_k$ be an interior visit in block B_i. There is an *entry* cross edge e_{in} of Q at u_1 and an *exit* cross edge e_{out} at u_k. By Lemma 6.2, e_{in} attaches at a terminal role r at u_1 with color $\text{pc}(r)$, and e_{out} at a role r' at u_k with color $\text{pc}(r')$. All edges of Q between vertices of W lie in B_i, so by Lemma 5.2(a) W is a PC path in H_{π_i}. If $k \geq 2$: the PC condition at u_1 (between e_{in} and the first edge of W) forces the first edge of W to have color $\neq \text{pc}(r)$, similarly at u_k; so W with roles r, r' is admissible of type (i). If $k = 1$: the PC condition at u_1 between e_{in} and e_{out} forces $\text{pc}(r) \neq \text{pc}(r')$: admissible of type (ii).

(c) A visit lies in one block. $\square$

Theorem 6.4 (Inflation Lemma). *For $a \geq 2$,*

$$\rho\big(\tau[\pi_1, \dots, \pi_a]\big) \ \leq \ (2a-3) \max_i M(\pi_i) \ + \ 2 \max_i b_i.$$

Proof. Let Q attain ρ with q visits. If $q = 1$, then $\rho \leq \max_i b_i$. If $q \geq 2$, then by Lemma 6.3 the $q - 2 \leq 2a - 3$ interior visits have at most $\max_i M(\pi_i)$ vertices each and the two end visits at most $\max_i b_i$ each; visits partition the vertices of Q. $\square$

Theorem 6.4 holds for *every* outer permutation τ. The problem is thereby reduced to finding patterns with small transit number relative to their size.

7 The gadget: classification of admissible visits of H_{T_b}

Definition 7.1. For even $b \geq 6$ let $T_b \in S_b$ be given by

$$T_b(0) = 0, \qquad T_b(1) = b-1, \qquad T_b(i) = i \ \ (2 \leq i \leq b-2), \qquad T_b(b-1) = 1.$$

Thus T_b^{-1} fixes 0 and each $v \in [2, b-2]$ and swaps $1 \leftrightarrow b-1$ (T_b is an involution). Recall $b \geq 6$.

Lemma 7.2 (Inventory)**.** *The edges of H_{T_b} are exactly:*

- doubled pairs $\{i, i+1\}$ *for* $2 \le i \le b-3$ *(each carrying one P- and one V-edge);*
- P-only edges $\{0,1\}$*,* $\{1,2\}$*,* $\{b-2, b-1\}$*;*
- V-only edges $\{0, b-1\}$*,* $\{2, b-1\}$*,* $\{1, b-2\}$*.*

The full edge lists at the relevant vertices are:

$$\begin{array}{ll} \textit{vertex } 0: & \mathrm{P}\,\{0,1\};\quad \mathrm{V}\,\{0,b{-}1\} \quad (\textit{degree } 2),\\ \textit{vertex } 1: & \mathrm{P}\,\{0,1\},\ \mathrm{P}\,\{1,2\};\quad \mathrm{V}\,\{1,b{-}2\} \quad (\textit{unique V-edge}),\\ \textit{vertex } b{-}1: & \mathrm{P}\,\{b{-}2,b{-}1\};\quad \mathrm{V}\,\{0,b{-}1\},\ \mathrm{V}\,\{2,b{-}1\} \quad (\textit{unique P-edge}),\\ \textit{vertex } 2: & \mathrm{P}\,\{1,2\};\ \textit{doubled } \{2,3\};\ \mathrm{V}\,\{2,b{-}1\},\\ \textit{vertex } b{-}2: & \textit{doubled } \{b{-}3,b{-}2\};\ \mathrm{P}\,\{b{-}2,b{-}1\};\ \mathrm{V}\,\{1,b{-}2\},\\ \textit{vertex } i,\ 3 \le i \le b-3: & \textit{doubled } \{i{-}1,i\} \textit{ and } \{i,i{+}1\} \textit{ only.} \end{array}$$

The terminals are $p_0 = v_0 =$ *vertex* 0*,* $v_1 =$ *vertex* 1*,* $p_1 =$ *vertex* $b-1$*.*

Proof. P-edges are $\{i, i+1\}$, $0 \le i \le b-2$. The V-edges, indexed by $v = 0, \dots, b-2$, are $\{T_b^{-1}(v), T_b^{-1}(v+1)\}$:

$$v=0: \ \{0,b{-}1\}; \qquad v=1: \ \{b{-}1,2\}; \qquad 2 \le v \le b-3: \ \{v,v{+}1\}; \qquad v=b{-}2: \ \{b{-}2,1\}.$$

A pair $\{i, i+1\}$ is doubled iff it appears in both lists: exactly for $2 \le i \le b-3$. For $b \ge 6$: $\{0,1\}$ doubled would need $b-1=1$; $\{1,2\}$ would need $b-3=1$; $\{b-2,b-1\}$ would need $b-3=1$; all false, and the three long V-edges join non-adjacent positions since $b-1, b-3 \ge 3$. The per-vertex lists and the terminals ($T_b^{-1}(0)=0$, $T_b^{-1}(b-1)=1$) follow by inspection. □

Let $S = \{2, 3, \dots, b-2\}$ (the *segment*), $|S| = b-3$. By Lemma 7.2, the sub-multigraph on S is the doubled path $2 - 3 - \cdots - (b-2)$; the only edges from S to its complement are the two at vertex 2 (P $\{1,2\}$, V $\{2, b-1\}$) and the two at vertex $b-2$ (P $\{b-2, b-1\}$, V $\{1, b-2\}$); and no terminal lies in S.

Lemma 7.3 (Segment behavior)**.** *Let Q be a PC path in H_{T_b}.*

(a) While inside S, Q moves monotonically in position (each interior vertex of S is adjacent only to its two position-neighbors), and Q can leave S only from vertex 2 or vertex $b-2$.

(b) Suppose Q enters S at one end (2 or $b-2$) by an edge of color c, at a moment when no vertex of S has yet been visited, and later leaves S. Then either it leaves immediately from the entry vertex (using the other external edge there, of color $\bar{c}$), or it traverses all of S to the far end; in the latter case its $b-4$ internal edges alternate colors starting with $\bar{c}$, so (since b is even, making $b-4$ even) the last internal edge has color c, and any edge by which Q then leaves S has color $\bar{c}$.

Proof. (a) is immediate from the inventory: a simple path cannot reverse inside a path graph. For (b): leaving from the entry vertex means the run in S is that single vertex; the two external edges at 2 (resp. $b-2$) have different colors, and the PC condition forces the exit edge to differ from the entry edge, i.e. to be the other one, of color $\bar{c}$. Otherwise the run proceeds monotonically and exits only at the far end, having covered all of S with internal edges $g_1, \dots, g_{b-4}$. The PC condition at the entry vertex gives $g_1 = \bar{c}$ (both colors are available on every doubled pair; the PC condition forces the alternation), alternation gives $g_m = \bar{c}$ for odd m and $g_m = c$ for even m, and $b-4$ is even, so $g_{b-4} = c$; the PC condition at the far end forces the exit edge to have color $\neq c$. □

7.1 The classification theorem

Theorem 7.4 (Classification of admissible visits)**.** *Let $b \geq 6$ be even. Up to reversal (Lemma 5.5), the admissible visits of H_{T_b} are exactly the following five, with their (unique) role pairs:*

	vertices	*edges (colors)*	*role pair*
(V1)	0	—	$\{p_0, v_0\}$
(V2)	$0,\ b-1$	$\mathrm{V}\{0, b{-}1\}$	$\{p_0, p_1\}$
(V3)	$0,\ 1$	$\mathrm{P}\{0, 1\}$	$\{v_0, v_1\}$
(V4)	$b-1,\ 0,\ 1$	$\mathrm{V}\{0, b{-}1\},\ \mathrm{P}\{0, 1\}$	$\{p_1, v_1\}$
(V5)	$b-1,\ 2,\ 1$	$\mathrm{V}\{2, b{-}1\},\ \mathrm{P}\{1, 2\}$	$\{p_1, v_1\}$

In particular: the role pairs $\{p_0, v_1\}$ and $\{v_0, p_1\}$ admit no visit; the only visit for $\{p_0, p_1\}$ is (V2); the only visit for $\{v_0, v_1\}$ is (V3); and $M(T_b) = 3$.

Proof. That each listed visit is admissible is a direct check from Lemma 7.2. (V1): vertex 0 carries p_0 and v_0 with different port colors. (V2): first = last edge $\mathrm{V} \neq \mathrm{pc}(p_0) = \mathrm{pc}(p_1) = \mathrm{P}$. (V3): first = last edge $\mathrm{P} \neq \mathrm{pc}(v_0) = \mathrm{pc}(v_1) = \mathrm{V}$. (V4), (V5): first edge $\mathrm{V} \neq \mathrm{pc}(p_1)$, last edge $\mathrm{P} \neq \mathrm{pc}(v_1)$, and the middle vertex alternates the colors. The role pairs are forced: e.g. in (V2) the first edge is V, so the role at 0 has port color $\neq$ V, i.e. p_0; the same one-line check fixes each role pair.

It remains to show no other admissible visit exists. Type-(ii) visits need a vertex carrying two roles of different port colors; by Lemma 7.2 the only such vertex is $0 = p_0 = v_0$, giving (V1). A type-(i) visit starts at a terminal vertex with a constrained first color (Lemma 5.5(b)). The terminal vertices are $0, 1, b-1$, giving exactly four (start vertex, first color) *stems*:

(A) $(0, \mathrm{V})$ [role p_0], (B) $(0, \mathrm{P})$ [role v_0], (C) $(1, \mathrm{P})$ [role v_1], (D) $(b-1, \mathrm{V})$ [role p_1].

Every admissible visit, read from its first vertex, extends one of these stems, so it suffices to enumerate, for each stem, every PC continuation, recording the moments at which the path stands at a terminal vertex with a last-edge color admissible for a role there. We use the *forcing facts* from Lemma 7.2: (F1) vertex 0 has exactly one P-edge ($\{0,1\}$) and one V-edge ($\{0, b-1\}$); (F2) vertex 1 has exactly one V-edge ($\{1, b-2\}$); (F3) vertex $b-1$ has exactly one P-edge ($\{b-2, b-1\}$); plus the segment behavior of Lemma 7.3. No vertex of S is a terminal, so a path stranded in S can never end an admissible visit there.

Stem (A): start 0, first edge V. By (F1) the first edge is $\mathrm{V}\{0, b-1\}$; the path stands at $b-1$, last color V. *Record:* role p_1 at $b-1$ has $\mathrm{pc} = \mathrm{P} \neq \mathrm{V}$, so $(0, b-1)$ is admissible: this is (V2). Continue: next $\neq$ V, so by (F3) $\mathrm{P}\{b-2, b-1\}$; enter S at $b-2$, entry color P, S untouched. By Lemma 7.3(b):

- *Immediate exit at $b-2$ by $\mathrm{V}\{1, b-2\}$:* stand at 1, last color V; the only role at 1 is v_1 ($\mathrm{pc} = \mathrm{V}$): not admissible. Continue from 1: next $\neq$ V, so a P-edge; $\{0, 1\}$ excluded (0 visited), leaving $\mathrm{P}\{1, 2\}$ into S at 2. From 2, next $\neq$ P: $\mathrm{V}\{2, b{-}1\}$ excluded ($b{-}1$ visited), leaving the V-side of $\{2,3\}$; from 3 the path moves monotonically up and strands (exits are 2, behind it, and $b-2$, visited). No admissible visit.

- *Full crossing to* 2: arrival at 2 by color P (Lemma 7.3(b) with $c=\mathrm{P}$); exit must be V; the only V-external edge at 2 is $\{2,b-1\}$ with $b-1$ visited. Strands. No admissible visit.

Stem (A) yields exactly (V2).

Stem (B): start 0**, first edge** P**.** By (F1) the first edge is P$\{0,1\}$; stand at 1, last color P. *Record:* role v_1 at 1 has pc $=\mathrm{V}\neq\mathrm{P}$: admissible: (V3). Continue: next $\neq$ P, so by (F2) V$\{1,b-2\}$; enter S at $b-2$, entry color V, S untouched. By Lemma 7.3(b):

- *Immediate exit at* $b-2$ *by* P$\{b-2,b-1\}$: stand at $b-1$, last color P; the only role at $b-1$ is p_1 (pc $=$ P): not admissible. Continue from $b-1$: next $\neq$ P, so a V-edge; $\{0,b-1\}$ excluded (0 visited), leaving V$\{2,b-1\}$ into S at 2. From 2, next $\neq$ V: P$\{1,2\}$ excluded (1 visited), leaving the P-side of $\{2,3\}$; monotone up and strands ($b-2$ visited). No admissible visit.

- *Full crossing to* 2: arrival color V; exit must be P; only P-external edge at 2 is $\{1,2\}$, and 1 is visited. Strands. No admissible visit.

Stem (B) yields exactly (V3).

Stem (C): start 1**, first edge** P**.** The P-edges at 1 are $\{0,1\}$ and $\{1,2\}$; two branches.

- *Branch* $1\to 0$. At 0, last color P. *Record:* v_0 (pc $=$ V) admissible: (V3) reversed. Continue: next $\neq$ P, by (F1) V$\{0,b-1\}$; at $b-1$, last color V. *Record:* p_1 admissible: (V4) reversed. Continue: next $\neq$ V, by (F3) P$\{b-2,b-1\}$ into S at $b-2$, entry color P, S untouched. Immediate exit by V$\{1,b-2\}$: 1 visited. Full crossing: arrival at 2 color P, exit must be V: only $\{2,b-1\}$, visited. Strands; branch closed.

- *Branch* $1\to 2$. Enter S at 2 by P$\{1,2\}$, entry color P, S untouched. Immediate exit by V$\{2,b-1\}$: at $b-1$, last color V. *Record:* p_1 admissible: (V5) reversed. Continue from $b-1$: next $\neq$ V, by (F3) P$\{b-2,b-1\}$ into S at $b-2$ – but S is now touched (vertex 2): from $b-2$ the non-P continuations are V$\{1,b-2\}$ (1 visited) and the V-side of $\{b-3,b-2\}$, after which the path moves monotonically down and strands at 3 (2 visited). Branch closed. Alternatively, full crossing $2\to b-2$ with entry color P arrives at $b-2$ by color P; exit must be V: only $\{1,b-2\}$, 1 visited. Strands; branch closed.

Stem (C) yields exactly (V3), (V4), (V5) (reversed).

Stem (D): start $b-1$**, first edge** V**.** The V-edges at $b-1$ are $\{0,b-1\}$ and $\{2,b-1\}$; two branches.

- *Branch* $b-1\to 0$. At 0, last color V. *Record:* p_0 admissible: (V2) reversed. Continue: next $\neq$ V, by (F1) P$\{0,1\}$; at 1, last color P. *Record:* v_1 admissible: (V4). Continue: next $\neq$ P, by (F2) V$\{1,b-2\}$ into S at $b-2$, entry color V, S untouched. Immediate exit by P$\{b-2,b-1\}$: $b-1$ visited. Full crossing: arrival at 2 color V, exit must be P: only $\{1,2\}$, 1 visited. Strands; branch closed.

- *Branch* $b-1\to 2$. Enter S at 2 by V$\{2,b-1\}$, entry color V, S untouched. Immediate exit by P$\{1,2\}$: at 1, last color P. *Record:* v_1 admissible: (V5). Continue from 1: next $\neq$ P, by

(F2) V$\{1, b-2\}$ into S at $b-2-S$ touched (vertex 2): from $b-2$ the non-V continuations are P$\{b-2, b-1\}$ ($b-1$ visited) and the P-side of $\{b-3, b-2\}$, then monotone descent stranding at 3 (2 visited). Branch closed. Alternatively, full crossing $2 \to b-2$ with entry color V arrives by color V; exit must be P: only $\{b-2, b-1\}$, $b-1$ visited. Strands; branch closed.

Stem (D) yields exactly (V2), (V4), (V5).

Collecting the four stems: every admissible visit is one of (V1)–(V5) up to reversal, each realized only with the listed role pair (checked at each *Record* step). No admissible visit realizes $\{p_0, v_1\}$ or $\{v_0, p_1\}$, and the maximum size is 3, attained by (V4) and (V5). □

Corollary 7.5. *$M(T_b) = 3$ for every even $b \geq 6$, and $c^* := \liminf_n \rho_{\min}(n)/n$ satisfies $c^* \leq 6/b$ for every even $b \geq 6$; hence $c^* = 0$ and the question of Section 1 has a negative answer.*

Proof. $M(T_b) = 3$ is part of Theorem 7.4. Taking $\pi_i = T_b$ for all i in Theorem 6.4 and letting $a \to \infty$: $\rho_{\min}(ab)/(ab) \leq \big((2a-3) \cdot 3 + 2b\big)/(ab) \to 6/b$. □

Remark 7.6 (The parity is essential)**.** For *odd* b, $b-4$ is odd, so the last internal edge of a full crossing has color $\bar{c}$ and the exit edge may have color c. The route $0 \xrightarrow{\text{P}} 1 \xrightarrow{\text{V}} b-2 \rightsquigarrow 2 \xrightarrow{\text{V}\{2,b-1\}} b-1$ (full crossing in the middle) then becomes properly colored, an admissible visit for $\{v_0, p_1\}$ of size b. Exact computation (Section 9) confirms $M(T_b) = b$ for odd $b \in \{7, 9, 11, 13\}$ and $M(T_b) = 3$ for even $6 \leq b \leq 100$, and confirms the *exact list* of Theorem 7.4 for even $6 \leq b \leq 14$ by independent brute-force enumeration (ancillary file `verify_M.py`).

Corollary 7.7 (Two visits to one T_b-block)**.** *Let $\sigma = \tau[\pi_1, \ldots, \pi_a]$ be an inflation, B_i a block whose pattern π_i equals T_b for some even $b \geq 6$, and Q a PC path in H_σ. Then the interior visits of Q to B_i have total size at most 4.*

Proof. By Lemma 6.2, B_i has at most four incident cross edges, one per role; each interior visit uses two distinct ones (entry and exit), and every cross edge is used at most once. Hence there are at most two interior visits; if at most one, its size is at most $M(T_b) = 3 \leq 4$. If two, they use all four cross edges, so their role pairs partition $\{p_0, p_1, v_0, v_1\}$, and they are vertex-disjoint (disjoint runs of the simple path Q). By Theorem 7.4 the three partitions give:

- $\{p_0, p_1\}/\{v_0, v_1\}$: the only visits are (V2)$= (0, b-1)$ and (V3)$= (0, 1)$, which share vertex 0 – impossible;
- $\{p_0, v_1\}/\{v_0, p_1\}$: no visits exist for either pair – impossible;
- $\{p_0, v_0\}/\{p_1, v_1\}$: the $\{p_0, v_0\}$ visit is (V1) at vertex 0, and the $\{p_1, v_1\}$ visit is (V4) or (V5); since (V4)$= (b-1, 0, 1)$ contains vertex 0 it cannot be disjoint from (V1), so the disjoint pair is (V1) with (V5), of sizes 1 and 3: total 4. (In every case the total is at most 4.)

□

8 Proof of the Main Theorem

Theorem 8.1. *(a) For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \leq 6a + 2b$, where $\sigma_{a,b} = \mathrm{id}_a[T_b, \ldots, T_b]$, i.e. $\sigma_{a,b}(jb + r) = jb + T_b(r)$.*

(b) For every $n \geq 100$, $\rho_{\min}(n) \leq 10\sqrt{n} + 10$.

In particular $\rho_{\min}(n) = O(\sqrt{n})$ and $\rho_{\min}(n)/n \to 0$, using only $M(T_b) = 3$ and the Inflation Lemma.

Proof. (a) Immediate from Theorem 6.4 with $M(\pi_i) = 3$ and $b_i = b$: $\rho \leq (2a-3)\cdot 3 + 2b \leq 6a+2b$.

(b) Let $b = 2\lfloor \sqrt{n}/2 \rfloor$, so b is even, $\sqrt{n} - 2 < b \leq \sqrt{n}$, and $b \geq 10$ since $n \geq 100$. Let $a = \lfloor n/b \rfloor \geq 2$ and $s = n - (a-1)b$, so $b \leq s < 2b$. Define $\sigma \in S_n$ as the inflation (identity outer permutation) of:

- if s even: $a - 1$ copies of T_b and one copy of T_s (s even, $s \geq b \geq 10$);
- if s odd: $a - 1$ copies of T_b, one copy of T_{s-3}, and one block of size 3 with identity pattern $\iota_3 = (0, 1, 2)$ ($s - 3$ even, $s - 3 \geq b - 2 \geq 8$, since s odd and b even force $s \geq b + 1$).

Every block has transit number at most 3: the T-blocks by Theorem 7.4, the ι_3-block trivially. The number of blocks is $m \leq a + 1 \leq n/b + 1$ and the largest block has size at most $2b - 1$. By Theorem 6.4,

$$\rho_{\min}(n) \;\leq\; (2m - 3) \cdot 3 + 2(2b - 1) \;\leq\; \frac{6n}{b} + 4b - 5.$$

Since $b \geq \sqrt{n} - 2$ and $\sqrt{n} \geq 10$,

$$\frac{6n}{b} \;\leq\; \frac{6n}{\sqrt{n} - 2} \;=\; 6\sqrt{n} + \frac{12\sqrt{n}}{\sqrt{n} - 2} \;\leq\; 6\sqrt{n} + 15,$$

using $\sqrt{n}/(\sqrt{n} - 2) \leq 10/8$ for $\sqrt{n} \geq 10$. With $4b \leq 4\sqrt{n}$ we get $\rho_{\min}(n) \leq 10\sqrt{n} + 10$. □

The refined constants use the classification once more, through Corollary 7.7.

Proposition 8.2 (Refined constants)**.** *For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \leq 4a + 2b$; and for every $n \geq 100$, $\rho_{\min}(n) \leq 8\sqrt{n} + 20$.*

Proof. First claim: let Q be a PC path in $H_{\sigma_{a,b}}$ with q visits. If $q = 1$ then $|Q| \leq b$. Otherwise, by Lemma 6.3 the first and last visits have at most b vertices each, and by Corollary 7.7 the interior visits contribute at most 4 per block, i.e. at most $4a$ in total. Hence $\rho \leq 4a + 2b$.

Second claim: take the construction from Theorem 8.1(b). Interior visits contribute at most 4 per T-block (Corollary 7.7; it applies verbatim to T_b, T_s, T_{s-3}) and at most $2 \cdot 3 = 6$ for the single ι_3-block (at most two interior visits – Lemma 6.2 – of size at most 3 each). End visits contribute at most $2(2b - 1)$. Hence

$$\rho_{\min}(n) \;\leq\; 4\Big(\frac{n}{b} + 1\Big) + 6 + 2(2b - 1) \;=\; \frac{4n}{b} + 4b + 8 \;\leq\; \big(4\sqrt{n} + 10\big) + 4\sqrt{n} + 8 \;\leq\; 8\sqrt{n} + 20,$$

using $4n/(\sqrt{n} - 2) = 4\sqrt{n} + 8\sqrt{n}/(\sqrt{n} - 2) \leq 4\sqrt{n} + 10$ for $\sqrt{n} \geq 10$. □

Proposition 8.3 (Explicit long path: the family is $\Theta(a + b)$)**.** *For every even $b \geq 6$ and $a \geq 2$, $\rho(\sigma_{a,b}) \geq 2a + 2b - 7$.*

Proof. Write (j, ℓ) for the vertex at position $jb + \ell$ ($0 \le j < a$, $0 \le \ell < b$), so block j is a copy of H_{T_b} on $\{(j, 0), \ldots, (j, b-1)\}$, with values $(j, 0) \mapsto jb$, $(j, 1) \mapsto jb + b - 1$, $(j, \ell) \mapsto jb + \ell$ for $2 \le \ell \le b - 2$, and $(j, b-1) \mapsto jb + 1$. Besides the internal edges (Lemma 7.2 in each block), the cross edges are the P-edges $\{(j, b-1), (j+1, 0)\}$ and the V-edges $\{(j, 1), (j+1, 0)\}$. Within block j note the V-edges $\{(j, 0), (j, b-1)\}$ and $\{(j, 2), (j, b-1)\}$.

The following walk is a PC path; colors are indicated over each step, and each edge exists by the inventory.

- *Stage 1 (block* 0*).* Start at $(0, b-2)$ and descend the segment to $(0, 2)$ along the doubled run, choosing colors V, P, V, …; this uses $b-4$ edges, and since $b-4$ is even the last (into $(0, 2)$) has color P. Then take V $\{(0, 2), (0, b-1)\}$, then the P-cross edge to $(1, 0)$.
- *Stage 2 (blocks* $1, \ldots, a-2$*).* From $(j, 0)$ (entered by a P-cross edge) take V $\{(j, 0), (j, b-1)\}$, then the P-cross edge to $(j+1, 0)$; repeat.
- *Stage 3 (block* $a-1$*).* From $(a-1, 0)$ (entered by a P-cross edge) take V $\{(a-1, 0), (a-1, b-1)\}$, then P $\{(a-1, b-2), (a-1, b-1)\}$, then descend the segment from $(a-1, b-2)$ to $(a-1, 2)$ choosing colors V, P, …, stopping at $(a-1, 2)$.

All vertices are distinct (stages use disjoint blocks; within a stage the listed vertices are distinct), and consecutive edge colors differ at every junction. The vertex count is $(b-2)$ in stage 1 ($b-3$ segment vertices plus $(0, b-1)$), $2(a-2)$ in stage 2, and $(b-1)$ in stage 3 ($(a-1, 0)$, $(a-1, b-1)$, and the $b-3$ segment vertices), totalling $2a + 2b - 7$. The path is checked mechanically, edge by edge, in the ancillary file `verify_path.py`. □

Remark 8.4 (Constants; exact values)**.** Together, Propositions 8.2 and 8.3 give $2(a+b) - 7 \le \rho(\sigma_{a,b}) \le 4a + 2b$: the family is $\Theta(a+b)$, and with $a \asymp b$ realizes $\Theta(\sqrt{n})$, so the upper bound of Theorem 1.1 is achieved by this family up to the constant. Exact machine computation (Section 9) gives $\rho(\sigma_{a,b}) = 2a + 2b - 4$ at every tested point up to $n = 720$; we conjecture this holds for all $a \ge 3$, even $b \ge 6$, but we neither prove nor use it. Optimizing the proven upper bound $4a + 2b$ at $n = ab$ with $b \asymp \sqrt{2n}$ gives $\rho_{\min}(n) \le 4\sqrt{2n} + O(1) \approx 5.66\sqrt{n}$ along such n.

9 Computational verification

None of the proofs above depend on computation. This section specifies the computations that independently verify them and the empirical sharpness of the construction. **All code, command lines, and output logs are provided as ancillary files with this submission** (directory `code/` in the repository, shipped as the standard ancillary directory `anc/` on arXiv; `code/README.md` maps each claim below to a script and a log, and `code/MANIFEST.sha256` lists SHA-256 checksums).

Algorithm. The basic object is an exact longest-PC-path search. Its state space is the set of triples (*visited set*, *current vertex*, *last edge color*); the initial states are all $(\{x, w\}, w, c)$ over edges $\{x, w\}$ of color c; a transition extends a state by an unvisited neighbor along an edge of the other color. Every PC path corresponds to a unique state, so $\rho(\sigma)$ is the maximum cardinality of the visited set over all *reachable* states, computed by an exhaustive search visiting every reachable state once (a memo set of states; no pruning). The method is exact by construction; its cost is the number of reachable states. That number is exponential in general – on unstructured instances we used the method only for $n \leq 19$ – but is small on the structured families here, because the gadget strands most partial paths quickly: for $\sigma_{a,b}$ we measure 57,066 reachable states at $n = 80$, 5,012,186 at $n = 600$, and 7,273,586 at $n = 720$ (the logs report the count for every table row). The constrained variants computing $M(\pi)$ (fixed start/end terminals and first/last colors) are the same search with the initial and accepting states restricted.

Implementations. Three independent implementations were used: a plain recursive enumerator and a memoized iterative one (both Python, in the artifact), which agree on 300 random permutations of sizes 4–8; and a separately written C++ solver (also in the artifact) from a parallel verification effort, which agrees with the Python solvers on every instance checked by both, including exact values of $\rho(\sigma_{a,b})$ at $(a, b) \in \{(6, 8), (5, 10), (4, 12)\}$ and the trap-family values of Appendix A at six points. In addition, the artifact contains a *minimal independent verifier* (`verify_M.py`, about 100 lines, written directly from Definition 5.4 with no shared code), which re-derives $M(T_b)$ and the complete list of admissible visits by brute force for $b \leq 14$.

Verified statements. Each item states the claim, its range, and its log.

1. **Anchors of Section 4** (`validate2.log`, `search_small.log`): $\rho(\sigma_m) = 2m + 3$ for the block-reversal family at $m = 1, 2, 3, 4$ ($n = 7, 11, 15, 19$); $\rho(\sigma^*) = 10$ for the $n = 18$ counterexample; $\rho_{\min}(7) = 5$ and $\rho_{\min}(8) = 6$ by full enumeration of S_7, S_8.

2. **Classification** (`verify_M.log`): for even $b \in \{6, 8, 10, 12, 14\}$ the admissible visits of H_{T_b} are *exactly* (V1)–(V5) of Theorem 7.4; for odd $b \in \{7, 9, 11\}$, $M(T_b) = b$.

3. **The family at scale** (`check_all_even.log`): $M(T_b) = 3$ for *every* even $6 \leq b \leq 100$; $M(T_b) = b$ for odd $b \in \{7, 9, 11, 13\}$ (`validate2.log`).

4. **Optimality of the gadget at small sizes** (`search_small.log`): full enumeration of S_b for $4 \leq b \leq 9$ gives $\min_{\pi \in S_b} M(\pi) = 3$ in every case; no gadget of size ≤ 9 beats transit number 3.

5. **Explicit lower-bound path** (`verify_path.log`): the path of Proposition 8.3 is checked edge-by-edge at seven (a, b) points including $(20, 20)$.

6. **Exact values** (`endtoend.log`, `bigtable.log`): for $\sigma_{a,b}$:

a	b	n	ρ (exact)	$2a+2b-4$	$4a+2b$	ρ/n
10	8	80	32	32	56	0.400
12	8	96	36	36	64	0.375
12	12	144	44	44	72	0.306
16	12	192	52	52	88	0.271
20	12	240	60	60	104	0.250
16	16	256	60	60	96	0.234
24	16	384	76	76	128	0.198
20	20	400	76	76	120	0.190
30	16	480	88	88	152	0.183
24	20	480	84	84	136	0.175
30	20	600	96	96	160	0.160
36	20	720	108	108	184	0.150

Every exact value equals $2a+2b-4$ and sits inside the proven window $[2a+2b-7,\ 4a+2b]$ of (1); the two rows at $n=480$ (differing in ρ) illustrate that $\rho(\sigma_{a,b})$ tracks a and b separately, not just their product. At $n=720$ the ratio 0.150 is far below any ratio that the block-reversal experiments of Section 4 suggested was attainable.

7. **Exhaustive small-n statistics** (`logs/exh7.txt`–`exh12.txt`, C++ solver): the joint distribution of $(b(\sigma),\rho(\sigma))$ over all of S_n for $7\le n\le 12$, used in Section 10; at $n=12$ this enumerates all 479,001,600 permutations.

8. **Landscape** (exploratory; `probe.py`): under adjacent transpositions, $\sigma_{4,6}$ ($n=24$, $\rho=16$) is a strict local minimum of ρ, while greedy descent from random starts stalls around $\rho\approx 23$ – extremal permutations of this type are found by design, not by local search.

Items 1–7 are exhaustive within their stated ranges and reproducible by running the listed scripts; item 8 is exploratory.

10 Lower bounds and open problems

We work with ±1-blocks (Definition 4.1); $b(\sigma)$ is their number and $s_{\max}(\sigma)$ the largest size. Two elementary facts frame the lower-bound question. First, the ±1-blocks partition the positions, so

$$s_{\max}(\sigma)\cdot b(\sigma)\ \ge\ n, \qquad \text{hence} \qquad \max\big(s_{\max}(\sigma),\, b(\sigma)\big)\ \ge\ \sqrt{n}.$$

Second, $\rho(\sigma)\ge s_{\max}(\sigma)$ always (Proposition 4.2). Consequently, if there were an absolute $c>0$ with $\rho(\sigma)\ge c\,b(\sigma)$ for all σ, then $\rho_{\min}(n)\ge c\sqrt{n}$ would follow, and with Theorem 8.1:

Conjecture 10.1. $\rho_{\min}(n)=\Theta(\sqrt{n})$.

We emphasize the asymmetry of the evidence: the upper bound is proved, while for the lower bound we have only the following partial information about ρ versus $b(\sigma)$, which we believe is of independent interest.

- **Exhaustively for $n\le 12$:** $\rho(\sigma)\ge b(\sigma)$ for every $\sigma\in S_n$ (machine-verified over all 479,001,600 permutations at $n=12$; logs `exh7.txt`–`exh12.txt`; equality occurs).

- **Asymptotically the constant 1 fails:** each copy of T_b contributes four ±1-blocks ($\{0\}$, $\{1\}$, $[2,b-2]$, $\{b-1\}$), so $b(\sigma_{a,b})=4a$, while exact computation (`bigtable.log`) gives $\rho(\sigma_{20,12})=60<80=b(\sigma_{20,12})$ at $n=240$, and similarly $76<96$ at $(24,16)$ and $96<120$ at $(30,20)$: finite, reproducible counterexamples to $\rho\ge b(\sigma)$.

- **Proven window:** by (1), $\frac{\rho(\sigma_{a,b})}{b(\sigma_{a,b})}\in\big[\frac12+\frac{2b-7}{4a},\ 1+\frac{b}{2a}\big]$; the conjectured exact formula $2a+2b-4$ would give limit $\frac12$ as $a/b\to\infty$. We know of no σ with $\rho(\sigma)<\frac12 b(\sigma)$.

Thus the sharp form of the lower-bound question is whether $\rho\ge c\,b(\sigma)$ holds for some $c\le\frac12$; any such c settles Conjecture 10.1. At the opposite extreme, even $\rho_{\min}(n)\to\infty$ appears open: the bound $\rho\ge s_{\max}+1$ degenerates for permutations with all blocks of bounded size, and our methods produce no lower bounds at all. (For the special case $b(\sigma)=n$ – no two consecutive positions carrying consecutive values – exhaustive computation for $n\le 12$ found every such permutation PC-Hamiltonian, $\rho=n$; this extreme case is wide open.)

On the gadget side, the exhaustive minimum $\min_{\pi \in S_b} M(\pi) = 3$ for $4 \le b \le 9$ hints that transit numbers cannot be made arbitrarily small; deciding whether $M(\pi) \ge 3$ for all $|\pi| \ge 4$, or the true minimum growth of M in b, would calibrate how far inflation alone can go. (Heuristically it cannot beat $\sqrt{n}$: the end-visit term ties the bound to the block size, and nesting inflations doubles the visit-count bound per level.)

Two further directions. First, the analogous question for the union of two Hamiltonian *cycles*: we expect the method to adapt – the port calculus acquires two wraparound cross edges and the visit count rises accordingly – but we have not carried out the analysis and do not claim the result. Second, the Lovász-side question (Section 1.3): for the graphs $G(n, \sigma)$ built from $\sigma_{a,b}$ or from the chains of Appendix A, is the maximum number of matching edges on a path still $\Theta(n)$? The properly-colored relaxation is now known to be lossy; quantifying the loss on these explicit instances may indicate whether $f(n)$ is linear after all. (An explicit snake path through $G(ab, \sigma_{a,b})$ using $\ge a(b-3) = n - 3a$ matching edges, checked edge-by-edge at six points up to $n = 600$ in `verify_f_linear.py`, already shows the loss is large on this family: the matching-edge content stays linear while ρ drops to $\sqrt{n}$.)

Acknowledgments

This paper merges and supersedes the author's two earlier notes on this problem – an extremal-conjecture note containing the conversion lemma, the block-reversal family, and the $n = 18$ refutation, and a second note proving the $O(\sqrt{n})$ bound – into one self-contained account. The proofs and computations were AI-assisted by Claude, under the Periapsis research structure. Because of that, I have tried to make the paper checkable without trusting either the author or the assistant. The central case analysis is stated as a complete classification (Theorem 7.4) whose every step is locally verifiable; every computational claim is reproducible from the ancillary artifact, which includes a minimal independent verifier written directly from the definitions; and the main theorem has a second, independent proof in Appendix A. This is an unrefereed preprint; no journal referee has yet checked it.

References

[1] L. Lovász, Problem 11, in: *Combinatorial Structures and their Applications* (Proc. Calgary Internat. Conf., 1969), Gordon and Breach, New York, 1970, p. 497.

[2] S. Norin, R. Steiner, S. Thomassé, and P. Wollan, *Long cycles in vertex-transitive graphs*, arXiv:2505.08634, 2025.

[3] X. Liu, X. Yu, and S. Zhang, *Circumference of 3-connected cubic graphs*, J. Combin. Theory Ser. B **132** (2018), 104–153.

[4] J. Bang-Jensen and G. Gutin, *Alternating cycles and paths in edge-coloured multigraphs: a survey*, Discrete Math. **165–166** (1997), 39–60.

[5] G. Gutin and E. J. Kim, *Properly coloured cycles and paths: results and open problems*, in: Graph Theory, Computational Intelligence and Thought, Lecture Notes in Comput. Sci. 5420, Springer, 2009, 200–208.

[6] A. Abouelaoualim, V. Borozan, Y. Manoussakis, C. Martinhon, and R. Muthu, *Properly colored paths in edge-colored graphs*, Discrete Math. **310** (2010), 1610–1620.

A A second construction: trap chains

This appendix gives an independent proof of $\rho_{\min}(n) = O(\sqrt{n})$ by a different family, found in a parallel verification effort. It does not use Sections 5–7; conversely, the main text does not use this appendix. The mechanism is again a parity obstruction, but localized differently: here *large reversed blocks of odd size, flanked by singletons*, are traps that no properly colored path can cross.

Definition A.1. Let $C = (c_0, c_1, \dots, c_r)$ be a composition of n with $c_t \geq 1$ and *no two consecutive entries equal to* 1. Put $P_t = c_0 + \cdots + c_{t-1}$, and let block t occupy positions and values $[P_t, P_t + c_t - 1]$, reversed:

$$\sigma_C(P_t + i) \;=\; P_t + c_t - 1 - i, \qquad 0 \leq i < c_t.$$

Write $L_t = P_t$ and $R_t = P_t + c_t - 1$ (the *ports* of block t).

Lemma A.2. *H_{σ_C} consists exactly of:*

(a) doubled pairs $\{p, p+1\}$ for all consecutive positions inside each block;

(b) for each $0 \leq t < r$, one P-only cross edge $\{R_t, L_{t+1}\}$;

(c) for each $0 \leq t < r$, one V-only cross edge $\{L_t, R_{t+1}\}$;

(d) nothing else.

*In particular the cross edges incident to block t attach as follows (*port–color crossover*): at L_t, a P-edge toward block $t-1$ and a V-edge toward block $t+1$; at R_t, a V-edge toward block $t-1$ and a P-edge toward block $t+1$ (omitting nonexistent neighbors). Non-port vertices carry no cross edges.*

Proof. (a) Inside a block, consecutive positions hold consecutive (descending) values: both a P-edge and a V-edge. (b) Positions R_t, L_{t+1} are consecutive; their values are P_t (block minimum) and $P_{t+1} + c_{t+1} - 1$ (block maximum), which differ by $c_t + c_{t+1} - 1 \geq 2$ since not both c_t, c_{t+1} equal 1: P-only. (c) The values $P_t + c_t - 1$ (maximum of block t, held at position L_t since the block descends) and $P_t + c_t = P_{t+1}$ (minimum of block $t+1$, held at R_{t+1}) are consecutive; the positions L_t, R_{t+1} differ by $c_t + c_{t+1} - 1 \geq 2$: V-only. (d) Positions in non-consecutive blocks are never adjacent; values in non-consecutive blocks never differ by 1 (value ranges are contiguous in block order); within consecutive blocks, (b)/(c) list the only position-adjacent (resp. value-adjacent) pairs not internal to a block. The crossover statement reads off (b) and (c) at blocks $t-1, t, t+1$. □

Lemma A.3. *Let Q be a PC path in H_{σ_C} and let t be a block containing no endpoint of Q. Then every maximal run of consecutive Q-vertices inside block t is either (i) a single port vertex, or (ii) a* full traversal *of all c_t vertices from one port to the other. In case (ii), if the entering cross edge has color χ, the exiting cross edge has color $\neq \chi$ if and only if c_t is odd.*

Proof. A run with no endpoint of Q has an entering and an exiting cross edge, which by Lemma A.2 attach only at ports; the run's internal edges lie in the block, whose internal graph is a doubled path. If the run enters and leaves at the same port it is that single vertex (re-reaching the entry port would repeat it); the two cross edges at one port have different colors (Lemma A.2), as the PC condition requires. Otherwise it goes from one port to the other through the entire internal path: a full traversal with $c_t - 1$ internal edges. The PC condition forces the internal colors to alternate starting with $\bar{\chi}$, so the last internal edge has color χ iff $c_t - 1$ is even; the exit edge differs from it, so it has color $\bar{\chi}$ iff c_t is odd. □

Lemma A.4 (Trap Lemma). *Let block t of σ_C have odd size $c_t \geq 3$ with singleton neighbors $c_{t-1} = c_{t+1} = 1$ $(1 \leq t \leq r-1)$. Let Q be a PC path with no endpoint in block t. Then Q uses at most 2 vertices of block t (at most one visit per port, and no full traversal).*

Proof. Write A for block t and s^-, s^+ for the single vertices of blocks $t-1$, $t+1$. By Lemma A.2 the four cross edges at A are

$$\text{at } L_A: \ \{s^-, L_A\} \text{ (P)}, \quad \{L_A, s^+\} \text{ (V)}; \qquad \text{at } R_A: \ \{s^-, R_A\} \text{ (V)}, \quad \{R_A, s^+\} \text{ (P)}.$$

Suppose Q makes a full traversal of A, entering by a cross edge of color χ; since c_t is odd it exits by a cross edge of color $\bar{\chi}$ (Lemma A.3). Four cases:

1. enter at L_A by $\{s^-, L_A\}$ ($\chi = \text{P}$): the exit is the V-colored cross edge at R_A, $\{s^-, R_A\}$ – but s^- was visited immediately before entering, a repetition;
2. enter at L_A by $\{L_A, s^+\}$ ($\chi = \text{V}$): the exit is $\{R_A, s^+\}$ – repeats s^+;
3. enter at R_A by $\{s^-, R_A\}$ ($\chi = \text{V}$): the exit is $\{s^-, L_A\}$ – repeats s^-;
4. enter at R_A by $\{R_A, s^+\}$ ($\chi = \text{P}$): the exit is $\{L_A, s^+\}$ – repeats s^+.

Each case contradicts simplicity of Q. So every visit is a single port vertex; there are two ports. □

Theorem A.5. *Let $L \geq 3$ be odd, $k \geq 1$, $0 \leq p \leq L+1$, and $C = (2, (L,1)^k, 3+p, 2)$, a composition of $n = (L+1)k + p + 7$ with no two consecutive 1's. Then*

$$\rho(\sigma_C) \ \leq \ 3k + 3L + p + 1.$$

Consequently $\rho_{\min}(n) \leq 7\sqrt{n} + O(1)$ for all sufficiently large n, by choosing L the largest odd integer $\leq \sqrt{n}$ and $k = \lfloor (n-7)/(L+1) \rfloor$, $p = n - 7 - (L+1)k$.

Proof. The blocks, in order, are: a 2-block; $A_1, s_1, A_2, s_2, \ldots, A_k, s_k$ where each A_t has size L and each s_t is a singleton; a $(3+p)$-block; a 2-block. For $2 \leq t \leq k$ the block A_t is flanked by the singletons s_{t-1} and s_t, so Lemma A.4 applies to it; A_1 is flanked on the left by the initial 2-block and is not trapped.

Fix a PC path Q and charge each block with the number of Q-vertices it contains; $\rho(Q)$ is the sum of the charges, bounded by category:

- each singleton s_t: charge ≤ 1; total $\leq k$;

- each trapped block A_t ($2 \le t \le k$) with no endpoint of Q: charge ≤ 2 (Lemma A.4); at most 2 trapped blocks may contain an endpoint of Q, each then with charge $\le L$ (its size). Hence the trapped blocks contribute at most $2(k-1) + 2(L-2)$ in total (each endpoint block exceeds the trap bound 2 by at most $L - 2$);

- A_1: charge $\le L$;

- the three remaining blocks: charges ≤ 2, $\le 3 + p$, ≤ 2; total $\le 7 + p$.

Summing: $\rho(Q) \le k + 2(k-1) + 2(L-2) + L + (7+p) = 3k + 3L + p + 1$. (If an endpoint lies in a non-trapped block, the corresponding excess term is not needed; the bound only improves.)

For the corollary: with $L \le \sqrt{n}$ odd and maximal, $k \le n/(L+1) \le n/L$ and $p \le L + 1$, so $\rho \le 3n/L + 4L + 4 = 7\sqrt{n} + O(1)$ using $L = \sqrt{n} - O(1)$. □

Remark A.6. The parity hypothesis is again essential: for even L, Lemma A.3 makes a full traversal *preserve* the boundary color, the four cases of Lemma A.4 become consistent, and computation shows the analogous even-L chains are fully traversable (ρ grows with slope 1). Exact computation (log `check_trap.log`, plus the parallel C++ solver) gives $\rho(\sigma_C) = 2k + 2(L-1)$ for $C = (2, (L, 1)^k, 3, 2)$, odd $L \in \{5, 7, 9, 13, 15, 21\}$ and all $k \ge 2$ within reach ($n \le 57$): the same per-period slope 2 as the family of Section 8. We do not know whether the two families are related beyond sharing this slope.